UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00971

LH RADIATION AND RELATIVITY

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. The a particle's 4-velocity, u^μ, and 4-acceleration, α^μ, are given by

$$u^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}, \qquad \alpha^\mu = \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = \frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2}.$$

(a) Explain what is meant by the *coordinate time*, t, and the *proper time*, τ, of the particle, and prove that $\mathrm{d}t = \gamma(v)\mathrm{d}\tau$, where $\boldsymbol{v}$ is the 3-velocity of the particle, and $\gamma(v)$ is the Lorentz factor. **[4]**

(b) Show that u^μ and α^μ can be written in terms of the 3-velocity, v, and 3-acceleration, $\boldsymbol{a}$, as **[6]**

$$u^\mu = \gamma(v)(c, \boldsymbol{v}), \qquad \alpha^\mu = \gamma^2(v)(0, \boldsymbol{a}) + \gamma^4(v)\frac{\boldsymbol{v}\cdot\boldsymbol{a}}{c^2}(c, \boldsymbol{v}).$$

(c) Consider the relativistic single harmonic oscillator for a particle of mass m with rectilinear motion occurring in the x-direction with $f_x = -\gamma(v)kx$ (as a component of a 3-vector), where k is a constant analogous to the spring constant. Show that **[10]**

$$\gamma^3(v)\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{k}{m}x.$$

(d) Derive an expression for $\mathrm{d}\gamma/\mathrm{d}x$. Combining this with the result of part (c), or otherwise, show that

$$\gamma(x) = \gamma(0) - \frac{kx^2}{2mc^2},$$

and explain the physical meaning of this result in terms of the energy of the particle. **[10]**

2. In this question we consider the features of electromagnetic field theory in $2+1$ dimensions. Greek indices $\alpha, \mu, \ldots$ denote $2+1$ spacetime indices $0, 1, 2$, and Roman indices $i, j, \ldots$ denote the 2-dimensional spacelike part, $1, 2$, so for instance the $2+1$ Minkowski metric tensor which is written $\eta^{\mu\nu}$ and $\varepsilon^{\alpha\beta\gamma}$ is the usual three-dimensional antisymmetric, Levi-Civita symbol.

   Throughout this question you may work in units where physical constants are unity, i.e. $\varepsilon_0 = \mu_0 = c = \hbar = 1$.

   Consider a real-valued $2+1$ vector potential field

$$A^\mu = (A^0, A^1, A^2) = (V, A_x, A_y),$$

   which satisfies the analogue of the Lorenz gauge equation $\partial_\mu A^\mu = 0$. An antisymmetric field tensor $F_{\mu\nu}$ is defined

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \begin{pmatrix} 0 & E_x & E_y \\ -E_x & 0 & -B \\ -E_y & B & 0 \end{pmatrix},$$

   involving an electric 2-vector (E_x, E_y) and magnetic 2-scalar B.

   (a) The relativistic form of Maxwell's equations in $2+1$-dimensional electromagnetism can be written

$$\Box^2 A^\nu = J^\nu.$$

   Using the Lorenz gauge expression, show that this is equivalent to

$$\partial_\mu F^{\mu\nu} = J^\nu,$$

   and thus show the components of the 2+1 Faraday tensor satisfy **[8]**

$$\partial_x E_x + \partial_y E_y = \rho, \qquad -\dot{E}_x + \partial_y B = J_x, \qquad -\dot{E}_y - \partial_x B = J_y.$$

   (b) Argue that the equation $\varepsilon^{\alpha\beta\gamma}\partial_\alpha\partial_\beta A_\gamma = 0$ is automatically satisfied, and show that this is equivalent to the equation **[6]**

$$\dot{B} + \partial_x E_y - \partial_y E_x = 0.$$

   (c) Calculate the 2+1 Lorentz scalar invariant $F^{\mu\nu}F_{\mu\nu}$. Can you construct a 2+1 analogue to the other electromagnetic scalar? **[4]**

(d) Now consider a 2+1 *Chern-Simons* vector potential $A^\mu = (V, A_x, A_y)$ satisfying the Lorenz gauge condition, but, instead of the stated inhomogeneous d'Alembert equation in (a), A^μ satisfies

$$2\varepsilon^{\alpha\beta\gamma}\partial_\beta A_\gamma = J^\alpha.$$

Show that this is equivalent to the following expressions in terms of the Chern-Simons electric field 2-vector (E_x, E_y) and magnetic field 2-scalar B:

$$B = -\tfrac{1}{2}\rho, \qquad E_x = -\tfrac{1}{2}J_y, \qquad E_y = \tfrac{1}{2}J_x.$$

Do the Chern-Simons fields satisfy any other Maxwell-like equations? **[6]**

(e) Assume now the source terms of the Chern-Simons field vanish, i.e. $\rho = J_x = J_y = 0$. Can you solve the Chern-Simons field equations in terms of E_x, E_y and B? How and why is this different from the more familiar $2+1$ Maxwell-like equations in part a) when the source terms vanish? **[6]**

3. (a) The equation of motion of a particle of charge q and rest mass m in an electromagnetic field can be formulated in relativistically covariant form as

$$\frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = \frac{q}{mc} F^{\mu\nu} p_\nu .$$

Deduce the temporal ($\mu = 0$) and spatial ($\mu \to i = (1, 2, 3)$) components of the equation of motion in terms of 3-vectors. Explain the physical meaning of your results. **[10]**

(b) An electron, charge e, rest mass m_e, crossing the laboratory at non-relativistic 3-velocity $\boldsymbol{v} = (v, 0, 0)$ enters the region of a uniform magnetic field $\boldsymbol{B} = (0, 0, B)$. Using the equation of motion and neglecting the radiation loss, show that the electron performs a semicircular U-turn, and leaves the region of the field after a time $\Delta t = \pi/\omega_B$ where $\omega_B \equiv eB/m_e c$ is the cyclotron frequency. Find the radius of the semicircle which constitutes the U-turn. **[10]**

(c) Using Larmor's formula

$$\frac{\mathrm{d}\mathcal{E}_{\mathrm{rad}}}{\mathrm{d}t} = -\frac{2e^2}{3c^3}\left(\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\right)^2 ,$$

show that the total radiation loss for the electron during the U-turn is given by **[5]**

$$|\Delta\mathcal{E}_{\mathrm{rad}}| = \frac{2\pi e^2 v^2 \omega_B}{3c^2} .$$

(d) Given that the cyclotron frequency ω_B does not exceed $10^{13}\mathrm{s}^{-1}$ in laboratory conditions, and the classical electron radius, $r_{\mathrm{cl}} \equiv e^2/m_e c^2 \approx 3\times10^{-13}\mathrm{cm}$, show that our neglect of the radiation loss in (b) was justified. **[5]**

Formula Sheet

- **Maxwell's equations** in 3-form, for electric field $\boldsymbol{E}$, magnetic field $\boldsymbol{B}$, electric current density $\boldsymbol{J}$ and charge density ρ, in empty space, with electric constant ε_0 and magnetic constant μ_0, where $c = 1/\sqrt{\varepsilon_0\mu_0}$,

$$\begin{aligned} \nabla\cdot\boldsymbol{B} &= 0, & \nabla\times\boldsymbol{E}+\dot{\boldsymbol{B}} &= 0, \\ \varepsilon_0\nabla\cdot\boldsymbol{E} &= \rho, & \mu_0^{-1}\nabla\times\boldsymbol{B}-\varepsilon_0\dot{\boldsymbol{E}} &= \boldsymbol{J}, \end{aligned}$$

- **Lorentz force equations** in in 3-form, for particle charge q, speed $\boldsymbol{v}$, momentum $\boldsymbol{p}$, and energy $\mathcal{E}$,

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}), \qquad \frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = q\boldsymbol{E}\cdot\boldsymbol{v}.$$

- **Lorentz transformed EM fields** after a Lorentz boost by velocity $\boldsymbol{v}$,

$$\begin{aligned} \overline{\boldsymbol{E}} &= \gamma(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{E}), \\ \overline{\boldsymbol{B}} &= \gamma(\boldsymbol{B}-c^{-2}\boldsymbol{v}\times\boldsymbol{E}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{B}) \end{aligned}$$

- **Electromagnetic potentials** with V the scalar potential, $\boldsymbol{A}$ the vector potential,

$$\boldsymbol{B} = \nabla\times\boldsymbol{A}, \qquad \boldsymbol{E} = -\dot{\boldsymbol{A}} - \nabla V.$$

- **Minkowski Metric**, given by $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$, where all components zero except

$$\eta_{00} = \eta^{00} = +1, \qquad \eta_{jj} = \eta^{jj} = -1, \qquad \text{for } j = 1,2,3.$$

- **Spacetime position** x^μ transforms as contravariant 4-vector,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{r}); \qquad x^\mu = (x_0, x_1, x_2, x_3) = (ct, -\boldsymbol{r}).$$

- **Covariant Derivative**, ∂_μ transforms as covariant 4-vector,

$$\partial_\mu = (\partial_0, \partial_1, \partial_2, \partial_3) = (c^{-1}\partial_t, \boldsymbol{\nabla}), \qquad \partial^\mu = (\partial^0, \partial^1, \partial^2, \partial^3) = (c^{-1}\partial_t, -\boldsymbol{\nabla}).$$

- **Lorentz transformation**, $\Lambda^{\overline{\mu}}_{\nu}$ from Σ to $\overline{\Sigma}$, for a boost by velocity $v = \beta c$ in standard configuration, has nonzero components

$$\Lambda^{\overline{0}}_{0} = \Lambda^{\overline{1}}_{1} = \gamma, \quad \Lambda^{\overline{0}}_{1} = \Lambda^{\overline{1}}_{0} = -\gamma\beta, \quad \Lambda^{\overline{2}}_{2} = \Lambda^{\overline{3}}_{3} = 1,$$

with the Lorentz factor $\gamma = \gamma(v) = (1-\beta^2)^{-1/2}$.

- **d'Alembertian operator**, a Lorentz scalar operator,

$$\Box^2 \;\equiv\; \eta^{\mu\nu}\partial_\mu\partial_\nu \;=\; c^{-2}\partial_t^2 - \nabla^2.$$

- **Faraday tensor** in matrix form

$$F_{\mu\nu} \;=\; \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

$$F^{\mu\nu} \;=\; \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

so $F_{0i} = E_i/c$ and $F^{ij} = -\varepsilon^{ijk}B_k$, for $i, j, k = 1, 2, 3$.
It follows from the 4-potential $A^\mu \;=\; (V/c, \boldsymbol{A})$ that

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- **dual Faraday tensor** $G_{\mu\nu}$ is given by

$$G_{\mu\nu} \;=\; \tfrac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta}, \qquad G^{\mu\nu} \;=\; \tfrac{1}{2}\varepsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}.$$

Its components satisfy $G_{0i} = -B_i$, $G^{ij} = -\varepsilon^{ijk}E_k$ for $i, j, k = 1, 2, 3$.

- **Levi-Civita symbol** in 3D satisfies

$$\varepsilon_{ijk} \;=\; \varepsilon^{ijk} \;=\; \begin{cases} +1 & \text{if } \; ijk \text{ is an even permutation} \\ -1 & \text{if } \; ijk \text{ is an odd permutation.} \end{cases}$$

Here, even permutations consist of $123, 231, 312$, and odd permutations are $132, 213, 321$. It also satifies $\varepsilon_{ijk}\varepsilon^{imn} = \delta_j^m\delta_k^n - \delta_k^m\delta_j^n$, and

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) \;=\; \varepsilon^{ijk}a_ib_jc_k.$$

- **Levi-Civita symbol** in 4D satisfies

$$\varepsilon_{\mu\nu\rho\sigma} \;=\; -\varepsilon^{\mu\nu\rho\sigma} \;=\; \begin{cases} +1 & \text{if } \; \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \; \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{if } \; \text{anything else.} \end{cases}$$

The signs of the permutations of $0, 1, 2, 3$ are given by

$+$	0123,	0231,	0312,	1032,	1203,	1320,	2013,	2130,	2301,	3021,	3102,	3210,
$-$	0132,	0213,	0321,	1023,	1230,	1302,	2031,	2103,	2310,	3012,	3120,	3201.

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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